The Unintended Consequences of Corporate Bond ETFs: Technical Specification for a Rebuild

Specification of Dannhauser and Hoseinzade, *Review of Financial Studies* 35 (2022), 51–90
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Specification only. No code, data or backtest. Written 1 November 2026.

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Start here

The result in one paragraph. The paper asks whether ETFs are a distinct source of fragility in corporate bonds. It compares two bonds of the same issuer that faced different ETF selling pressure in the 2013 Taper Tantrum. The more exposed bond widens more, by 8.5 to 11.3 bps across the four significant months in the LQD setting. The gap then fades by December 2013.

The rebuild is feasible but not fully determined. A quant can rebuild all the regressions from this document. Twelve choices still need a decision, and Section 7.1 ranks them. The first is the most serious, because the paper gives two different rules for picking the control bond.

You cannot run the core tests on the data here. This user holds TRACE for 15 issuers from 2023 to 2025. The core tests need all US corporate bond TRACE from April 2013 to May 2014. Section 2.4 lists what each table needs.

How to read the labels. Every specification item carries one of these labels.

- **[STATED p. N]**: the paper says it, on that page or table.
- **[INFERRED**: reason**]**: the paper does not say it, but I derive it from something it does say.
- **[MISSING**. Proposal: default**]**: the paper does not say it. The default is my proposal, not the paper's.
- **[MINE**: comment**]**: my own suggestion or view, never the authors'.

Page numbers. Page numbers refer to the advance-access PDF, which runs 1 to 40. **[INFERRED**: the issue runs 51 to 90 and has the same 40 pages, so issue page = PDF page + 50. I did not see the issue pagination.**]**

What I read.

- All 40 pages of the PDF, as text and as page images for every table and figure.
- The OUP supplementary file, `hhab031_supplementary_data.docx`. It holds Appendices A and B only. Its numbers match the PDF appendix in a token-by-token comparison. It adds no new method.
- The SSRN version (abstract 3559433). I did not read it. A search confirmed that it exists, at 62 pages, but the page refused the fetch with HTTP 403. It may hold detail that the journal version cut.

The paper mentions no separate internet appendix. The OUP article page, as far as I could see it, listed only that one file and no replication code. Pages: `https://academic.oup.com/rfs/article/35/1/51/6179821` and `https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3559433`.

1 Research question, hypotheses and identification

1.1 Research question

Are the design features of ETFs a unique source of fragility for corporate bonds? **[STATED p. 2]** The paper names three features: intraday exchange trading, in-kind creation and redemption, and a liquidity gap that widens under stress. **[STATED p. 2]**

1.2 Hypotheses

The paper states these in prose. The labels H1 to H6 are mine, for reference.

	Hypothesis	Test	Label
H1	ETFs attract investors with greater liquidity needs than mutual funds.	Equation (3), Table 2	**[STATED p. 2, p. 12]**
H2	Exchange trading enables positive feedback trading, so ETF flows respond more to lagged returns.	Equation (4), Table 3	**[STATED p. 2, p. 13–14]**
H3	ETFs hold smaller liquidity buffers than mutual funds.	Table 1, panel A	**[STATED p. 3, p. 16]**
H4	ETF outflows reach bonds more directly. The trade-to-flow slope is near one for ETFs and below one for active funds.	Equation (5), Table 4	**[STATED p. 3, p. 17]**

	Hypothesis	Test	Label
H5	Under stress, authorized participants (APs, the dealers who exchange baskets of bonds for ETF shares) keep the largest and most liquid ETFs near NAV through redemptions. This turns ETF selling into bond selling.	Figure 2, Tables 5, 6, A1	**[STATED p. 3–4, p. 21]**
H6	Redemptions create price pressure followed by reversal, not price discovery.	Equation (6), Tables 8–10	**[STATED p. 23, p. 31]**

1.3 Identification logic

The Taper Tantrum is the shock. **[STATED p. 3, p. 19]** The paper treats it as a fundamental shock that produced a correlated demand shock. All bonds face the fundamental shock. Only the ETF exposure differs across bonds of one issuer.

The design removes confounds in four steps. **[STATED p. 4, p. 23–24]**

1. **Issuer fixed effects** absorb any issuer-level change in credit risk during the episode.
2. **Restricting the sample to bonds in ETF portfolios** removes selection into the ETF universe.
3. **Measuring spreads against a maturity-matched Treasury** removes term structure differences.
4. **Issue-level controls and mutual fund pressure** address bond-level rate sensitivity and mutual fund selling.

The variation that identifies the effect in each setting

Setting	Identifying variation	Assumption it rests on	What it addresses
1. LQD threshold (Table 8)	Two bonds of one issuer on either side of the $750 million index size rule.	The size rule is the only thing that keeps control bonds out of LQD and its redemptions. **[STATED p. 4, p. 25]**	Selection into the most liquid ETF.
2. Trade-based pressure (Table 9)	Cross-bond differences in actual ETF holding changes, from funds in the extreme flow quartiles.	Within an issuer, ETF managers do not choose the bonds to sell on bond-level fundamentals. Setting 3 exists because this may fail. **[INFERRED**: p. 28 raises manager discretion as the concern that setting 3 answers**]**	Whether the bond was truly in the baskets. **[STATED p. 4, p. 26]**
3. Hypothetical arbitrage (Table 10)	Pre-episode portfolio weights multiplied by each fund's net creation and redemption dollars.	Funds scale all holdings in proportion. Pre-episode weights are unrelated to later bond-specific news. **[STATED p. 11, p. 28 for proportional scaling]** **[INFERRED**: the second part follows from using pre-period weights to avoid discretion**]**	Manager discretion in choosing baskets. **[STATED p. 4, p. 28]**

Pressure against price discovery. The test uses the path of coefficients. Price discovery would need negative and significant coefficients after September 2013, as exposed bonds return first. Or it would need control bonds to catch up. Pressure predicts positive coefficients that fade. **[STATED p. 31]** The paper rejects catch-up from panel B of Figure 3, which is a visual argument and not a formal test. **[STATED p. 31]**

Limit the authors set themselves. They do not rule out broad price discovery through ETFs. They do not compare ETF prices to NAV in this test, or ETF bonds to non-ETF bonds. **[STATED p. 5, p. 31]**

Dates of the episode. The paper uses several windows, so name the one you mean.

- Onset: 22 May 2013, the Bernanke testimony. **[STATED p. 19]**
- End of turmoil: 18 September 2013, the Fed decision to keep the programme. **[STATED p. 19]**
- LQD redemption statistic: 21 May to 18 September. **[STATED p. 23]**
- Flow window: May, June, July and August 2013. **[STATED p. 11]**
- Creation and redemption window: 22 May to 31 August. **[STATED p. 11]**
- Rating screen window: 21 May to 31 August. **[STATED p. 11]**

- Outcome months: August 2013 to May 2014. The start follows Cella, Ellul and Giannetti (2013), and two proxies use flows through August. **[STATED p. 25, fn. 27]**

2 Data

2.1 Sources, fields, frequency and filters

The overall sample runs from January 2010 to December 2018. **[STATED p. 6]** The event tests use 2013 and 2014 only.

Sources as used by the paper

Source	Fields	Frequency, period	Filters and labels
CRSP Survivor-Bias-Free US Mutual Fund Database	Fund identification, holdings, returns, total net assets (TNA), turnover, expense ratio, share class data, Lipper code, fund name.	Monthly returns and TNA. Quarterly holdings. 2010–2018. **[STATED p. 6, p. 11]**	Keep funds with average corporate bond weight above 20% and a taxable fixed-income Lipper asset code. **[STATED p. 6]** Keep $0.5 < \text{TNA}_t/\text{TNA}_{t-1} < 2$. **[STATED p. 8]** Name rules appear in Section 4. **[STATED p. 7, fn. 12]**
Morningstar Mutual Fund Database	Investment class weights, average credit quality, average effective duration, average coupon.	**[MISSING.** Proposal: monthly where available, carried forward within the quarter**]**	Matched to CRSP on fund CUSIP. **[STATED p. 6]**
CRSP ETF daily	Daily market price (`prc`) and daily NAV (`dnav`).	Daily. January 2012 to December 2014 in Figure 2. **[STATED p. 10, p. 22]**	**[MISSING.** Proposal: all ETFs in the fund sample with both fields present**]**
Bloomberg	Daily ETF shares outstanding (`shrout`).	Daily. **[STATED p. 10]** **[MISSING.** Proposal: cover 1 January 2013 to 30 June 2014, the widest test window**]**	None stated.
TRACE	Bond trades. The paper does not say whether it uses Standard or Enhanced TRACE. **[MISSING.** Proposal: Enhanced TRACE, because it shows true size above the dissemination caps**]**	Trade level. 2010–2018 for liquidity. May 2013 to May 2014 for spreads. **[INFERRED:** Tables 4 and 8–10 need these spans**]**	Dick-Nielsen (2009) error filter. Remove trades below $100,000 par. Median and reversal filters (see below). Remove bonds under 1 year to maturity. **[STATED p. 8]**
Mergent Fixed Income Securities Database (FISD)	Bond characteristics, ratings from three agencies, amount outstanding, issue events.	**[MISSING.** Proposal: reference data as of each date, with the amount history table**]**	Exclusions appear in Section 4. **[STATED p. 10–11]**
Treasury rates	Daily maturity-matched Treasury rate.	Daily. **[STATED p. 8]**	Vendor **[MISSING.** Proposal: Federal Reserve H.15 constant maturity yields, linearly interpolated to the bond's remaining maturity**]**

Source	Fields	Frequency, period	Filters and labels
iBoxx USD Liquid Investment Grade Index rules	$750 million minimum, 3 years minimum to maturity, issuer rules.	Rules as of 2013. **[STATED p. 25]**	The paper cites a Markit link for the issuer rules and does not list them. **[MISSING**. Proposal: obtain the 2013 rule book. Without it, apply only the size and maturity rules**]**
SIFMA and Federal Reserve Board	Market size, issuance, holdings by investor type, dealer inventory.	Annual, 1996–2018. **[STATED p. 6–7]**	Figure 1 only. It is context, not a test.

TRACE price filters. The median filter drops a trade whose price differs by more than 20% from the daily median or the 9-day median. The reversal filter drops a trade whose price moves more than 10% and then reverses. **[STATED p. 8, fn. 13]** **[MISSING**. Proposal: for the reversal, flag a trade when its absolute price change from the previous trade exceeds 10% and the next trade moves back in the opposite direction by more than 10%. Check this against Edwards, Harris and Piwowar (2007), which I did not read**]**

2.2 Reported sample counts

The paper gives no attrition table. It reports only the counts below. A step-by-step count of bonds through the filters is **[MISSING**. Proposal: produce your own attrition table at each step in Section 4, and treat it as a deliverable**]**

Item	Count	Source
Corporate bonds held by the funds before the onset	10,131	**[STATED p. 10]**
Share of these held by ETFs, active funds, index funds	67%, 96%, 48%	**[STATED p. 10]**
Distinct issuers in ETF holdings	1,491	**[STATED p. 10]**
ETF-holding issuers not held by active funds	68	**[STATED p. 10]**
Distinct index funds, and those with corporate weight over 50%	36 and 3	**[STATED p. 3, fn. 4]**
Funds in the cross-section, without and with controls	1,128 and 873	**[STATED Table 2, p. 13]** **[INFERRED**: each observation is one fund**]**
Share classes in the cross-section, without and with controls	2,540 and 2,145	**[STATED Table 2, p. 13]** **[INFERRED**: each observation is one share class**]**
LQD setting, bonds per monthly regression	151 to 171	**[STATED Table 8, p. 27]**
Trade-based setting, bonds per monthly regression	999 to 1,247	**[STATED Table 9, p. 29]**
Hypothetical setting, bonds per monthly regression	1,258 to 1,491	**[STATED Table 10, p. 30]**
Number of ETFs, overall and per tercile	not reported	**[MISSING**. Proposal: report your own. Do not treat it as a check**]**
Number of issuers in each setting	not reported	**[INFERRED**: about 75 to 85 in Table 8, if most issuers have one treated and one control bond**]**

2.3 Plausible equivalents

All of this subsection is **[MINE**: my suggestions, not the paper's. Field names below come from my memory of these vendors, so verify them**]**

Paper's source	Equivalent
CRSP mutual fund data	Morningstar Direct or LSEG Lipper for flows and TNA. SEC N-CSR and N-Q filings on EDGAR for 2013 holdings. N-PORT starts only in 2019 and cannot serve this window.
Morningstar portfolio statistics	Lipper fund data, or sponsor fact sheets for the few largest ETFs.

Paper's source	Equivalent
CRSP ETF price and NAV	Bloomberg `PX_LAST` and `FUND_NET_ASSET_VAL`. Sponsor premium and discount history, such as iShares downloads.
Bloomberg shares outstanding	Sponsor daily history, or the shares field in an ETF data vendor file. Public ICI data are aggregates and do not suffice.
Enhanced TRACE	WRDS Enhanced TRACE. The WRDS Bond Returns product gives monthly cleaned data, but its window will not match the last-three-days rule.
Mergent FISD	Bloomberg reference data, or LSEG reference data.
Treasury rates	FRED H.15. Gürkaynak, Sack and Wright zero curves as a robustness check.
iBoxx index rules and constituents	IHS Markit (now S&P Global) archive. LQD historical holdings from iShares as a check on eligibility.

2.4 What the study needs, and what this user lacks

This user cannot run the bond-level tests. TRACE here covers 15 issuers from 2023 to 2025, so the Taper Tantrum window is absent. The tests need the whole bond cross-section in 2013. **[MINE**: I assessed feasibility only from the description given to me, and did not query the warehouse**]**

Minimum data for each output. All rows are my derivation.

Output	Needs	TRACE needed?
Tables 1A, 2, 3	CRSP funds and Morningstar, December 2008 to December 2018. The 12-month volatility needs a run-up year, or the first values fall in 2011.	No
Table 5	CRSP monthly TNA and return, December 2012 to December 2013.	No
Figure 2, Tables 6, A1	CRSP daily ETF price and NAV, 2012 to 2014. Daily shares, 2013. ETF bid and ask. Holdings of March and September 2013.	No
Table 4, columns without controls	CRSP quarterly holdings and monthly TNA, 2010 to 2018.	No
Table 4, columns with controls	As above, plus FISD amount outstanding and all-market TRACE from 2009 for the lagged Amihud measure.	Yes, 2009–2018
Table 1B	Holdings at the last report before May 2013. FISD. TRACE around May 2013.	Yes, about 2013
Tables 7–10, B1–B4, Figure 3	TRACE from April 2013 to May 2014 (July 2014 for Figure 3), with January 2013 for Figure 3 panel A. FISD with ratings and events. Daily Treasury rates. Fund holdings from about March to September 2013. Shares from May to August 2013. Index rules.	Yes, all bonds

3 Variables

Subscripts: i bond, j issuer, f fund, t month, d day, q quarter. Winsorising: the paper states it for the bond spread only. **[STATED p. 8–10]** For every other variable, the paper is silent. **[MISSING**. Proposal: no winsorising, so that the winsorising step cannot explain any gap to the targets. Report a 1% and 99% version as a robustness check**]**

3.1 Fund flow and fund characteristics

Variable	Formula and units	Timing and label
$Flow_{f,t}$	$\frac{\text{TNA}_{f,t} - \text{TNA}_{f,t-1}(1+R_{f,t})}{\text{TNA}_{f,t-1}}$. Percent in tables. Eq. (1).	Monthly. **[STATED p. 8]** Dollar flow is the numerator, in $ millions. **[STATED Table 5, p. 20]** Fund-level returns are TNA-weighted across share classes. **[STATED p. 8]**
$Flow_{f,\text{Tantrum}}$	Geometric average of the four monthly flows. **[INFERRED**: monthly flows can be negative, so the only workable form is $[\prod_m(1+Flow_{f,m})]^{1/4}-1$**]**	May to August 2013. **[STATED p. 11]**
Flow volatility	Standard deviation of monthly flow over 12 months. Percent.	Rolling monthly in the panel. Average per fund in the cross-section. **[STATED p. 12–13]** Rolling window **[MISSING**. Proposal: trailing 12 months with all 12 observations required**]**
Expense ratio, turnover	CRSP. Percent in Table 1. **[STATED p. 9]**	Lagged 1 month in panels. Averaged and demeaned in the cross-section. **[STATED p. 12–13]** Regression scale **[MISSING**. Proposal: percent, as in Table 1**]**
log(age)	Natural log of years since inception. **[INFERRED**: the text says log of age in years, and the paper does not state the base**]**	Inception date **[MISSING**. Proposal: CRSP first offer date of the oldest share class**]**
log(assets)	Log of TNA.	As for expense ratio. **[STATED p. 12]**
log(family assets)	Log of total fund family assets. **[STATED p. 12]**	Family definition **[MISSING**. Proposal: sum of TNA across CRSP funds with one management company code**]**
Credit quality	Portfolio average, numeric. **[INFERRED**: lower means safer, because p. 8 says index funds have higher quality and they show the lowest score**]**	Morningstar. **[STATED p. 6]** Coding scale **[MISSING**. Proposal: use the same rating scale as for bonds, and check against the Table 1 quartiles**]**
Effective duration, coupon	Portfolio average. Years and percent. **[STATED Table 1, p. 9]**	Lagged 1 month in Table 3. **[STATED p. 14]**
% corporate, cash, Treasuries	Portfolio weights. Percent. **[STATED Table 1, p. 9]**	Morningstar. The "average corporate weight" screen uses a period **[MISSING**. Proposal: average over all fund months in 2010–2018**]**
ETF_f, $Index_f$	Dummies from CRSP identifiers and name strings. **[STATED p. 6–7, fn. 12]**	Fixed by fund. Vanguard ETF share classes form separate funds. **[STATED p. 6–8]**

3.2 ETF arbitrage variables

Variable	Formula and units	Timing and label
Deviation, daily	$(\text{prc}_{f,t,d} - \text{dnav}_{f,t,d})/\text{dnav}_{f,t,d}$. Percent in tables. Eq. (3). Positive means premium.	Daily. **[STATED p. 10]**
C/R dollars, daily	$\text{dnav}_{f,t,d}\,(\text{shrout}_{f,t,d} - \text{shrout}_{f,t,d-1})$. Dollars. Eq. (4). Positive means net creation.	Daily. **[STATED p. 10]**
C/R dollars, Tantrum	Sum of daily values. Dollars.	22 May to 31 August 2013. **[STATED p. 11]** A version with lagged NAV gives the same results. **[STATED fn. 31, p. 28]**
Redemption dollars	Monthly sum of daily C/R on days when the shares change is negative.	**[STATED Figure 2, p. 22]**
Deviation terciles	Sort ETFs by average daily deviation in the Tantrum. Bottom: Discount. Middle: Low Deviation. Top: Premium. **[STATED p. 21–22]**	Window **[MISSING**. Proposal: 22 May to 31 August 2013**]**

Variable	Formula and units	Timing and label
Net C/R quintiles	Sort ETFs each month by total C/R dollars. **[STATED Table A1, p. 33]**	Monthly, May to August 2013. Average volatility is the volatility of daily deviation. Scale **[MISSING**. Proposal: report in the same units as deviation and compare ratios, since the Table A1 scale is unclear**]**
ETF bid-ask	(ask − bid) / midpoint, end of day, averaged. Percent. **[STATED Table 6, p. 23]**	January to April, and May to August 2013.
Volume, ETF turnover, size	Secondary volume, turnover, total assets. **[STATED Table 6, p. 23]**	The row says shares and the note says dollars. **[INFERRED**: use shares, because the row values look like daily share counts for ETFs of this size**]**

3.3 Pressure measures

ETF pressure, Eq. (5), then multiplied by minus one. **[STATED p. 11]**

$$ETFpressure_{ij} = -1 \times \left[\frac{\sum_f \max(0, \Delta H_{ijf} \mid F_f > P_{75})}{AO_{ij}} - \frac{\sum_f \max(0, -\Delta H_{ijf} \mid F_f < P_{25})}{AO_{ij}} \right] \tag{1}$$

- ΔH_{ijf} is the change in holdings, from the report before May to the first report after August 2013. **[STATED p. 11]**
- F_f is $Flow_{f,\text{Tantrum}}$. P_{25} and P_{75} are its quartiles. **[STATED p. 11]**
- Positive values mean net selling pressure. **[INFERRED**: the raw term is buys minus sells, and the paper flips the sign "to aid in interpretation"**]**
- Units are percent of amount outstanding. **[INFERRED**: p. 26 reads the slope per 1%, and Table 7 means are 0.277 and 0.458**]**
- The numerator and denominator are both par. **[MISSING**. Proposal: use reported par from holdings. If only market value exists, divide by price at the report date**]**
- Quartiles use ETFs only for the ETF measure. **[INFERRED**: p. 11 computes the measure "for ETFs", then a "similar measure for all mutual funds"**]**
- AO_{ij} is dated $t-1$ in the paper. **[INFERRED**: the date hardly matters, because the paper drops bonds whose amount outstanding changes**]**
- A bond with no trade by an extreme-quartile fund gets zero, or may leave the sample. The paper says the measure is "defined for bonds in the ETF universe that are in baskets". **[STATED p. 26]** **[INFERRED**: Table 9 has fewer bonds than Table 10, so bonds with no ETF trade probably leave the sample**]** **[MISSING**. Proposal: keep a bond in Table 9 only if at least one ETF changed its holding between the two reports. Test the alternative that keeps zeros**]**

MF pressure. The same measure for all mutual funds, active and index combined, also times minus one. **[STATED p. 11, Table 8 note]** **[INFERRED**: quartiles are computed across all mutual funds pooled, since the paper describes one combined measure**]**

ETF hypothetical arbitrage exposure, Eq. (6), then multiplied by minus one. **[STATED p. 11–12]**

$$ETFArb_{ij} = -1 \times \sum_{f=1}^{F} \frac{w_{ijf,\text{pre}} \times C/R\,\text{dollars}_{f,\text{Tantrum}}}{AO_{ij}} \tag{2}$$

- $w_{ijf,\text{pre}}$ is the weight of the bond in fund f at the holdings report before the episode. **[STATED p. 11]** **[MISSING**. Proposal: market value divided by fund TNA, as a fraction**]**
- Units are percent of amount outstanding. **[INFERRED**: p. 28 reads the slope per 1%. So multiply by 100 if the weight is a fraction**]**
- The numerator is at market value and the denominator at par. **[INFERRED**: this follows from the two formulas. The paper does not correct for it**]**
- The paper reports no summary statistics for this measure. **[MISSING**. Proposal: report mean, standard deviation and quartiles, because you cannot check the scale otherwise**]**
- The paper says the scaling removes the mechanical link that Wardlaw (2020) describes. **[STATED p. 12]**

LQD bond dummy. One for treatment bonds, zero for control bonds. **[STATED p. 25]**

3.4 Bond spreads and liquidity

Variable	Formula and units	Timing and label
$Spread_{ij,t}$	Volume-weighted yield minus the daily maturity-matched Treasury rate. Percentage points. **[INFERRED**: a coefficient of 0.113 is read as 11.3 bps on p. 26**]**	All trades in the last three full trading days of month t. **[STATED p. 8, p. 25]** The table notes say "business days", and fn. 14 settles it: half days do not count. **[STATED fn. 14, p. 8]**
Order of operations	**[MISSING**. Proposal: compute a spread for each trade as yield minus the Treasury rate of that day, then take the par-volume-weighted mean**]**	Yield source **[MISSING**. Proposal: the yield in Enhanced TRACE if present. Otherwise, compute yield to maturity from price and FISD terms**]**
Winsorising	1% and 99%. **[STATED p. 8–10]** **[INFERRED**: applied to the spread level, because the paper says the "spread values" are winsorised**]**	Cross-section or pooled **[MISSING**. Proposal: by month, over all bonds that pass the filters, before differencing**]**
$Spread_{ij,\text{May}}$	As above.	Three days before the testimony. **[STATED p. 25]** **[INFERRED**: 17, 20 and 21 May 2013, because 22 May was a Wednesday and 18–19 May a weekend**]**
Spread change	$Spread_{ij,t} - Spread_{ij,\text{May}}$. Percentage points.	t = August 2013 to May 2014. **[STATED p. 25]**
Amihud_t	Monthly median of the daily value $\frac{1}{N_d}\sum_d \frac{\lvert r_d\rvert}{v_d}$. Eq. (2). r_d: return between consecutive transactions. v_d: dollar volume in millions.	**[STATED p. 10]** Which trade gives v_d **[MISSING**. Proposal: the later trade of each pair**]** Minimum trades **[MISSING**. Proposal: two trades a day**]**
Amihud scale	Percent per $ million in Table 1 (mean 0.73). Fraction per $ million in regressions. **[INFERRED**: Table 7 shows 0.011 and 0.009. A slope of about 11 on 0.73 would imply 8 percentage points, which is not plausible**]**	**[MINE**: treat this as a mandatory first scale check**]**
Amihud in Table 4	Quarterly lagged 6-month moving average, divided by its standard deviation. **[STATED Table 4 note, p. 18]**	Which standard deviation **[MISSING**. Proposal: cross-sectional, within quarter**]**
Roll	$2\sqrt{-\text{cov}(R_t, R_{t-1})}$, using trades in the past 21 trading days, with at least four observations. Discard a positive covariance.	**[STATED p. 33]** Regression scale **[INFERRED**: fraction, for the same reason as Amihud**]**
IRC	$(P_{\max} - P_{\min})/P_{\max}$ within an imputed round trip, which is a set of same-size trades on the same day.	**[STATED p. 33]** Monthly aggregate **[MISSING**. Proposal: monthly mean. Weight round trips equally**]**
Turnover (Table 7)	Used but not defined. Values are 0.049 and 0.041.	**[MISSING**. Proposal: monthly traded par divided by amount outstanding, as a fraction**]**

3.5 Bond controls

The controls in Equation (6) are set to their May levels. **[STATED p. 25]**

Variable	Definition	Label
Time to maturity	Years to maturity.	**[STATED p. 25]** Date **[MISSING**. Proposal: 21 May 2013**]**
Coupon	Percent.	**[STATED p. 25]**
log(size)	Log of amount outstanding.	**[STATED p. 25]** Base **[MISSING**. Proposal: natural log, in dollars. Units affect only the intercept**]**
Option dummy	One if the bond has option features.	**[STATED p. 25]** FISD fields **[MISSING**. Proposal: one if the bond is callable or putable**]**. Make-whole only calls are a known issue. **[MISSING**. Proposal: count them as callable, then test without them**]**

Variable	Definition	Label
Amihud	May level.	**[STATED p. 25]** Window **[MISSING**. Proposal: 1 to 21 May 2013, to stay before the onset. Test the full month as well**]**
May spread	The base spread.	**[STATED p. 25]**
Rating	Numeric. AAA = 1, AA+ = 2, AA = 3, AA− = 4, and so on.	**[STATED Table 1 note, p. 9]** Combining agencies **[MISSING**. Proposal: mean of the most recent ratings available, rounded only for display, following fn. 15's remark about average ratings**]**
Ownership (Table 1)	Percent of the bond held by each fund type.	**[INFERRED**: the row and the panel B note imply this**]**
Own (Table 4)	Previous ownership of the bond by the fund.	**[STATED Table 4 note]** Scale **[MISSING**. Proposal: percent of amount outstanding**]**

3.6 Trade variable for Table 4

$Trade_{ijf,q}$ is the percent change in holdings of bond i by fund f between reports in $q-1$ and q. **[STATED p. 16]**

$Flow_{f,q}$ is the quarterly flow. **[STATED p. 16]** **[MISSING**. Proposal: apply Equation (1) to TNA between the two report dates**]**

The slope is tested against one, so the two must share units. **[INFERRED**: a test of $\beta = 1$ is only meaningful if both are percent**]**

A new position has no defined percent change. **[MISSING**. Proposal: drop new positions, since the paper's new-bond statistic is an unreported test in fn. 20. A full exit counts as −100%**]**

4 Sample construction, in order

The paper does not give an order. It lists the rules. **[INFERRED**: the order below is mine, chosen so that each step has its inputs ready**]**

A. Funds

1. Start from all CRSP funds, January 2010 to December 2018. **[STATED p. 6]**
2. Keep funds with a taxable fixed-income Lipper asset code and an average corporate bond weight above 20%. **[STATED p. 6]**
3. Delete exchange-traded notes, leveraged and inverse funds. Their lower-case names contain any of: plus, enhanced, inverse, 2x, 3x, ultra, 1.5x, 2.5x, exchange traded note, exchanged-traded note. **[STATED fn. 12, p. 7]**
4. Label ETFs. Use CRSP identifiers, and names with ishares, spdr, holdrs, streetracks, exchange traded or exchange-traded. **[STATED fn. 12, p. 7]**
5. Label index mutual funds among the remainder. Use CRSP identifiers, and names with index, idx or indx. **[STATED fn. 12, p. 7]**
6. Split Vanguard funds into a separate ETF fund and index fund. **[STATED p. 6–8]**
7. Merge Morningstar portfolio data on fund CUSIP. **[STATED p. 6]**
8. Aggregate to portfolio level. Sum share class assets and value-weight the other characteristics. **[STATED p. 8]**
9. Compute monthly flow. Drop observations with TNA ratio outside (0.5, 2). **[STATED p. 8]** The paper says this removes "funds". **[MISSING**. Proposal: drop the fund-month only. A permanent drop would remove the fund's entire history**]**
10. For share class regressions only, drop share classes and ETFs under $10 million and retirement share classes. **[STATED fn. 17, p. 12]** **[MISSING**. Proposal: identify retirement classes with the CRSP retirement flag, or an R in the class name**]**

B. Bond trades and reference data

11. Clean TRACE with the Dick-Nielsen (2009) filter. **[STATED p. 8]**
12. Remove trades under $100,000 par. **[STATED p. 8]**

13. Apply the median and reversal filters. **[STATED p. 8]**
14. Remove bonds with under 1 year to maturity. **[STATED p. 8]**
15. Merge FISD on bond CUSIP and build numeric ratings. **[STATED p. 10]** **[INFERRED**: the merge key is the standard one, and the paper does not name it**]**
16. Compute spreads, winsorise, then compute Amihud, Roll and IRC. **[STATED p. 8–10, p. 33]**

C. Event sample

17. Remove bonds with under 1 year and 4 months to maturity. **[STATED p. 11]** Date **[MISSING**. Proposal: measure at 21 May 2013**]**
18. Remove bonds whose rating from any of the three agencies crosses investment grade between 21 May and 31 August 2013. **[STATED p. 11]**
19. Remove sinking fund, defaulted, convertible, foreign currency, floating-rate and Yankee bonds. **[STATED p. 11]**
20. Remove bonds with amount outstanding changes: partial call, conversion, repurchase, pay-in-kind, tender or exchange. **[STATED p. 11, fn. 16]** Window **[MISSING**. Proposal: 22 May 2013 to 31 May 2014, because the outcome runs to May 2014. Test a window that ends on 31 August 2013**]**
21. Keep only bonds in ETF portfolios at the last report before May 2013. **[STATED p. 4, p. 24, p. 28]**

D. Exposures

22. Compute Tantrum flow for each fund. Find quartiles within ETFs and within mutual funds. **[STATED p. 11]**
23. Compute ΔH across the two reports, then ETF pressure and MF pressure. Flip the sign. **[STATED p. 11]**
24. Sum daily C/R dollars from 22 May to 31 August. Compute ETFArb. Flip the sign. **[STATED p. 11–12]**

E. The three regression samples

25. **Setting 1, LQD.** Require at least 4 years to maturity before the Taper Tantrum, for both groups. **[STATED p. 25]**
26. Treatment: within each issuer, take the smallest bond at or above $750 million. If two tie, keep both. **[STATED p. 25, fn. 29]** The bond must be investment grade. **[STATED p. 25]** The paper does not say whether the bond must truly be in LQD at the report date. The text and the Table 8 note say "meets the eligibility requirements". **[MISSING**. Proposal: use eligibility as the main rule, then a version that also requires an actual LQD holding**]**
27. Control: within the same issuer, a bond below $750 million, held by another ETF. **[STATED p. 25]** The paper then gives two rules for picking it. See the first row of the ranked table in Section 7.1. **[MISSING**. Proposal: closest May Amihud, one control per treatment, without replacement**]**
28. The alternative matches use Roll and IRC (Table B1), and time to maturity (Table B2). **[STATED p. 26]**
29. **Setting 2, trade-based.** All event-sample bonds with a defined ETF pressure. **[STATED p. 26]**
30. **Setting 3, hypothetical.** All event-sample bonds held by ETFs. **[STATED p. 28]**
31. For each month t, keep bonds with a May spread and a spread in month t. **[INFERRED**: the bond count changes each month, so the sample is not balanced**]** Keep issuers with two or more bonds in that month. **[INFERRED**: singletons add nothing under issuer fixed effects, and their presence does not affect the slope estimates, but it does affect N**]** **[MISSING**. Proposal: report N both ways**]**

5 Regressions and tests

Standard errors in all panels are clustered, and t-statistics appear in the tables. Stars: * $p < .1$, ** $p < .05$, *** $p < .01$. **[STATED table notes]**

5.1 Flow volatility (Table 2)

$$\mathrm{SD}_{f,t} = \alpha + \beta_1 ETF_f + \beta_2 Index_f + \gamma' X_{f,t} + \varepsilon_{f,t} \tag{3}$$

- The paper writes β_2 twice in Eq. (7). I use γ for the controls. **[INFERRED**: a typographical slip, since the terms differ**]**
- Controls: expense ratio, turnover, log age, log TNA, log family assets. **[STATED p. 12]**
- Cross-section (columns 1, 2, 5, 6): dependent variable is the fund's average 12-month volatility. Controls are averages over the sample, then demeaned. No clustering or fixed effects are reported. **[STATED p. 12–13]** **[MISSING**. Proposal: use heteroskedasticity-robust standard errors**]**

- Panel (columns 3, 4, 7, 8): monthly volatility, controls lagged 1 month, month fixed effects, clustered by fund or share class. **[STATED p. 13]**
- Columns 1–4 use funds. Columns 5–8 use share classes. **[STATED p. 13]**
- Columns 1, 3, 5, 7 have only the ETF dummy. **[STATED Table 2]**

5.2 Flow-performance sensitivity (Table 3)

$$Flow_{f,t} = \lambda_t + \delta(Ret_{f,t-1} \times ETF_f) + \beta_1 Ret_{f,t-1} + \beta_2 ETF_f + \beta_3' X_{f,t-1} + \varepsilon_{f,t} \tag{4}$$

- Coefficient of interest: δ. **[STATED p. 14]**
- Month fixed effects. Clusters by fund or share class. **[STATED p. 14–15]**
- Columns 1 and 5: no controls. Columns 2 and 6 add duration, credit quality, log size, log age, expense ratio, turnover. Columns 3 and 7 add three lags of flow. Columns 4 and 8 add 12-month flow volatility. **[STATED Table 3]**
- The comparison group is all mutual funds, active and index. **[INFERRED**: the note defines the dummy against "mutual funds", and no index dummy appears**]**

5.3 Trade response to flows (Table 4)

$$Trade_{ijf,q} = \alpha + \tau_q + \beta_1 Flow_{f,q} + \beta_2' X_{ijf,q} + \varepsilon_{ijf,q} \tag{5}$$

- Run separately for ETFs, active funds and index funds, and separately for outflows and inflows. That gives 12 columns. **[STATED p. 16–18]**
- Controls: lagged Amihud, ownership, and each interacted with flow. **[STATED p. 16–17, Table 4]**
- Quarter fixed effects. Clusters by fund. **[STATED Table 4 note]**
- Test: $H_0 : \beta_1 = 1$. The table gives p-values. **[STATED Table 4]**
- Outflow means $Flow < 0$ and inflow means $Flow > 0$. **[INFERRED**: the words imply this. The paper does not state how it treats zero flow**]**
- **[MINE**: with the interaction terms, β_1 is the slope at Amihud = 0 and Own = 0. It is not the slope at the mean, because the paper scales Amihud without saying it is demeaned. Compare the two before reading 0.946 as the average response**]**

5.4 Descriptive evidence on ETF arbitrage

Output	Construction	Label
Figure 1	Market size and issuance. Holdings by investor type and dealer inventory. Annual.	**[STATED p. 6–7]**. Context only.
Table 1	Panel A: fund characteristics, 2010–2018. Panel B: bond holdings at the report before onset, by fund type.	**[STATED p. 9]**
Table 5	Monthly distribution of flows in 2013 by fund type. Percent and $ millions.	**[STATED p. 20]**
Figure 2	Panel A: mean deviation by tercile, monthly, January 2012 to December 2014. Panel B: total assets by tercile, May to August 2013. Panel C: total redemption dollars by tercile, same months.	**[STATED p. 22]**. I read the axis of panel A, and it covers the caption's dates, not those on p. 21.
Table 6	Discount against Low Deviation terciles. January to April, and May to August 2013. Means and p-values of differences.	**[STATED p. 23]** Test **[MISSING**. Proposal: two-sample t-test with unequal variances**]**
Table A1	Monthly quintiles on net C/R, May to August 2013.	**[STATED p. 33]**
Table 7	Treatment against control bond means, with t-values and p-values.	**[STATED p. 26]** Test **[MISSING**. Proposal: two-sample t-test**]**. Date **[MISSING**. Proposal: May levels, as for the regression controls**]**

Output	Construction	Label
Figure 3	Panel A: mean spread difference of LQD held against same-issuer not held. Panel B: mean spreads. January 2013 to July 2014.	**[STATED p. 24]**. Holdings date **[MISSING**. Proposal: last report before May 2013**]**. Weights **[MISSING**. Proposal: equal-weighted bonds**]**. Sample **[INFERRED**: the caption says bonds held by LQD, which is wider than the matched sample in Table 8**]**

5.5 The main event regressions (Tables 8, 9, 10)

All three settings are one equation, Eq. (10). **[STATED p. 25]** Each runs as a separate cross-section per month, for ten months. **[STATED p. 24–25, table headers]**

$$\left(Spread_t - Spread_{\text{May}}\right)_{ij} = \lambda_j + \beta_1 ETFproxy_{ij} + \beta_2 MFpressure_{ij} + \beta_3' X_{ij,\text{May}} + \varepsilon_{ij,t} \tag{6}$$

for $t \in \{\text{Aug 2013}, \ldots, \text{May 2014}\}$. The notes to Tables 9 and 10 add α, which is only a normalisation alongside λ_j. **[INFERRED**: an intercept is collinear with a full set of issuer effects**]**

The three versions of Equation (6)

	Setting 1: Table 8	Setting 2: Table 9	Setting 3: Table 10
ETFproxy	LQD bond dummy	ETF pressure, Eq. (1)	ETFArb, Eq. (2)
MF pressure	Included	Included	Excluded **[STATED p. 28]**
Controls X	Time to maturity, coupon, Amihud, May spread. No option dummy because it has no variation. No size because it correlates with treatment. **[STATED p. 26]**	Time to maturity, coupon, log size, option dummy, Amihud, May spread. **[STATED Table 9]**	Same as setting 2. **[STATED Table 10]**
Fixed effects	Issuer	Issuer	Issuer
Clustering	Issuer **[STATED Table 8]**	Issuer **[STATED Table 9]**	Issuer **[STATED Table 10]**
Bond sample	Matched treatment and control	Bonds with a defined ETF pressure	All ETF-held event bonds
Robustness	Table B1 (match on Roll, IRC), Table B2 (match on maturity, with each proxy)	Table B3 (Roll, IRC as the proxy in X)	Table B4 (Roll, IRC as the proxy in X)

Economic scaling. The paper reports 20.35 bps as the peak price effect of a 1% pressure rise, "accounting for average duration". **[STATED p. 26–28]** **[INFERRED**: $0.045 \times 100 \times 4.52 = 20.34$ bps, using the peak Table 9 coefficient and the ETF mean effective duration in Table 1. The fourth digit may come from an unrounded input**]**

Subsamples. The paper splits funds by type and flow sign in Table 4, and bonds by setting. It splits the main regressions by no other dimension. It does not split them by rating, maturity or liquidity. **[INFERRED**: no such table appears**]**

No pre-period test. The paper runs no placebo regression before the event. **[INFERRED**: no such table appears**]** See Section 8.

6 Target numbers

All numbers below come from the PDF. A script parsed most tables from the PDF text and confirmed that every number occurs there. I also compared the rendered tables with the page images. The tables keep the paper's numbers, so Table 8 here is Table 8 there. T-statistics are in parentheses. The paper reports no standard errors. **[MINE**: derive them as $|\text{coefficient}/t|$. For example, the Table 8 August standard error is $0.113/2.67 = 0.042$**]**

Table 1, p. 9. Means and medians. The paper also gives SD and quartiles.

	ETFs		Active funds		Index funds	
	Mean	50%	Mean	50%	Mean	50%
A. Funds						
Flow monthly (%)	1.92	0.13	0.00	−0.23	0.36	0.31
Return monthly (%)	0.20	0.16	0.19	0.19	0.12	0.09
Total net assets ($ bln)	3.11	0.36	2.08	0.47	10.65	1.70
Expense ratio (%)	0.35	0.30	0.90	0.91	0.32	0.26
Turnover (%)	45.70	25.00	114.82	70.00	102.42	77.00
Age (years)	4.72	4.26	8.46	6.00	7.72	5.90
Credit quality	8.76	9.00	10.23	9.00	4.26	3.00
Effective duration (years)	4.52	4.39	4.07	4.24	5.78	5.53
Coupon (%)	4.48	4.31	4.51	4.18	3.38	3.28
% in corporate bonds	73.88	87.62	56.09	51.04	37.99	26.38
% in cash	2.83	0.91	4.88	3.77	2.77	1.33
% in Treasuries	12.48	2.13	17.17	11.24	38.42	42.06
12m flow vol, fund	7.13	5.14	3.75	2.26	3.69	2.27
12m flow vol, share class			4.88	2.70	4.38	2.56
B. Bonds held, report before onset						
Spread (%)	1.66	1.25	2.24	1.39	1.26	1.12
Size ($ mln)	728.30	500.00	637.67	499.98	734.32	500.00
Coupon (%)	5.33	5.50	5.66	5.75	4.91	5.14
Time to maturity (years)	9.62	6.20	9.00	5.91	10.54	6.82
Rating	8.88	8.00	9.75	9.00	7.64	7.00
Amihud	0.73	0.43	0.85	0.47	0.78	0.47
Roll	0.45	0.34	1.04	0.36	0.49	0.37
Imputed round-trip cost	0.21	0.17	0.25	0.18	0.21	0.17
Ownership (%)	2.21	1.99	9.47	6.14	2.00	1.82

Table 2, p. 13. Flow volatility. Columns 1–4 funds, 5–8 share classes. Columns 1, 2, 5, 6 cross-section.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ETF	3.312***	1.454***	3.361***	2.155***	2.048***	0.908*	1.374***	0.871*
	(10.19)	(2.59)	(9.36)	(4.60)	(5.14)	(1.73)	(3.74)	(1.91)
Index		0.983*		0.523		−0.344		−0.089
		(1.92)		(1.51)		(−0.84)		(−0.26)
Expense ratio		−0.014		−0.005		−0.004		0.003
		(−0.78)		(−0.27)		(−0.16)		(0.20)
Turnover ratio		−0.017		−0.003		0.025		0.004
		(−1.01)		(−0.41)		(1.36)		(0.76)
log(age)		−0.842***		−0.411***		−1.879***		−1.768***
		(−6.59)		(−6.96)		(−17.44)		(−19.22)
log(assets)		−0.571***		−0.686***		−0.266***		−0.459***
		(−9.36)		(−12.48)		(−6.03)		(−15.51)
log(family assets)		0.149***		0.182***		0.007		0.032
		(3.04)		(3.59)		(0.20)		(0.95)
Date fixed effects	No	No	Yes	Yes	No	No	Yes	Yes
Observations	1,128	873	102,266	73,376	2,540	2,145	267,410	230,584
Adj. R^2	.084	.234	.049	.117	.009	.173	.013	.121

Table 3, p. 15. Flow-performance sensitivity. Columns 1–4 funds, 5–8 share classes. Date fixed effects in all.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ret(t-1) x ETF	0.844***	0.565***	0.475***	0.484***	0.769***	0.570***	0.516***	0.524***
	(6.21)	(3.45)	(2.71)	(2.76)	(5.86)	(4.05)	(3.37)	(3.43)
Ret(t-1)	0.265***	0.335***	0.240***	0.241***	0.347***	0.401***	0.267***	0.266***
	(5.71)	(7.31)	(5.66)	(5.65)	(11.26)	(12.22)	(8.65)	(8.51)
ETF	2.467***	1.885***	0.986***	0.575***	2.004***	0.324	0.259*	0.048
	(10.25)	(7.82)	(6.42)	(3.69)	(9.88)	(1.50)	(1.81)	(0.31)
Effective duration		−0.096***	−0.053***	−0.042***		−0.010	−0.003	−0.001
		(−3.82)	(−3.78)	(−2.93)		(−0.61)	(−0.29)	(−0.06)
Credit quality		−0.022*	−0.016**	−0.022***		−0.025***	−0.022***	−0.032***
		(−1.68)	(−2.12)	(−2.82)		(−2.70)	(−3.89)	(−5.49)
log(size)		0.078***	0.041**	0.109***		0.205***	0.111***	0.160***
		(2.87)	(2.47)	(6.60)		(10.74)	(9.13)	(13.16)
log(age)		−0.303***	−0.154***	−0.119***		−1.516***	−0.777***	−0.591***
		(−7.50)	(−7.00)	(−5.22)		(−23.31)	(−18.66)	(−14.73)
Expense ratio		0.003	0.001	−0.001		0.009**	0.004*	0.002
		(0.89)	(0.49)	(−0.26)		(2.57)	(1.81)	(0.84)
Turnover		−0.009	−0.004	−0.005		−0.002	0.001	−0.000
		(−1.19)	(−1.04)	(−1.11)		(−0.49)	(0.60)	(−0.20)
Flow(t-1)			0.169***	0.159***			0.168***	0.155***
			(13.79)	(13.02)			(23.43)	(21.40)
Flow(t-2)			0.136***	0.126***			0.111***	0.097***
			(11.09)	(10.23)			(16.27)	(14.17)
Flow(t-3)			0.118***	0.106***			0.108***	0.092***
			(14.02)	(12.59)			(18.88)	(15.94)
Vol flow				0.137***				0.156***
				(9.81)				(18.09)
Observations	97,822	58,828	58,045	58,045	204,536	140,257	133,966	133,966
Adj. R^2	.033	.034	.122	.130	.017	.042	.110	.121

Table 4, p. 18. Trade response to flows. Quarter fixed effects in all. Clustered by fund.

	ETFs				Active funds				Index funds			
	Out	Out	In	In	Out	Out	In	In	Out	Out	In	In
Flow	0.872***	0.946***	0.362***	0.411***	0.426***	0.440***	0.216***	0.267***	0.153	0.156	0.103*	0.113
	(15.75)	(15.13)	(8.92)	(8.24)	(6.20)	(4.94)	(8.02)	(8.90)	(1.37)	(1.29)	(1.73)	(1.54)
Amihud		−0.466**		−0.452*		0.663***		0.584***		0.002		−0.307***
		(−2.16)		(−1.82)		(5.04)		(4.94)		(0.02)		(−3.47)
Own		−3.159***		−4.289***		0.827***		0.685***		−3.978*		−0.312
		(−3.15)		(−5.94)		(3.62)		(4.47)		(−1.82)		(−0.83)
Amihud x Flow		−0.054***		−0.036**		−0.013		−0.014		0.014		0.002
		(−3.57)		(−2.22)		(−0.76)		(−1.45)		(0.95)		(0.17)
Own x Flow		0.143		0.108**		0.007		−0.042***		−0.091		−0.009
		(1.36)		(2.10)		(0.19)		(−4.67)		(−0.21)		(−0.15)
Obs.	179,191	142,330	346,560	278,521	1,628,939	1,017,761	1,618,023	1,011,777	217,522	155,510	423,827	303,088
Adj. R^2	.186	.168	.124	.104	.114	.065	.105	.055	.091	.111	.091	.095
p [$\beta = 1$]	.02	.39	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00

Table 5, p. 20. Monthly flows in 2013. $ rows are $ millions.

	ETFs				Active funds				Index funds			
	Mean	p10	p50	p90	Mean	p10	p50	p90	Mean	p10	p50	p90
Jan %	7.46	−0.44	2.17	20.96	1.83	−2.02	0.40	7.11	−0.63	−6.21	0.10	5.20
Jan $	30.82	−0.18	3.33	162.53	20.99	−13.76	1.08	65.32	−20.55	−64.21	0.69	48.25
Feb %	6.79	−2.33	2.45	21.04	1.02	−2.09	0.10	5.26	1.05	−2.90	0.99	5.02
Feb $	12.66	−18.30	5.00	168.56	11.39	−16.92	0.09	47.38	186.39	−112.78	3.61	83.91
Mar %	4.36	−1.19	1.47	14.50	1.31	−2.22	0.27	5.72	0.82	−2.24	0.54	3.34
Mar $	40.34	−13.05	2.99	136.25	12.22	−15.92	0.37	43.86	56.36	−51.79	3.56	58.10
Apr %	2.81	−1.18	−0.02	11.79	1.19	−2.11	0.23	6.34	1.02	−1.10	0.40	3.64
Apr $	41.05	−15.93	−0.01	289.38	9.04	−15.57	0.37	42.69	65.86	−9.06	4.57	99.78
May %	6.00	−0.84	2.72	15.72	1.50	−2.67	−0.04	5.60	0.66	−2.08	0.50	5.40
May $	24.92	−4.40	5.02	129.67	10.12	−22.16	−0.02	47.66	−255.91	−33.98	1.76	101.07
Jun %	4.10	−7.82	−0.21	17.49	−1.05	−6.59	−1.08	3.74	0.08	−2.64	−0.30	2.83
Jun $	−54.08	−89.58	−0.04	34.15	−41.85	−119.81	−2.32	15.66	−59.08	−214.86	−2.05	81.74
Jul %	3.07	−7.66	−0.10	19.99	0.68	−4.04	−0.37	6.66	−0.43	−4.26	0.02	3.71
Jul $	30.01	−38.94	−0.02	128.74	−11.53	−46.23	−0.52	46.48	−8.16	−192.72	0.55	75.13
Aug %	2.56	−6.37	−0.15	12.60	0.18	−4.76	−0.82	5.44	0.94	−2.44	−0.09	4.69
Aug $	−23.51	−107.48	−0.04	56.56	−13.67	−59.10	−1.58	41.84	81.30	−30.93	−0.70	138.94
Sep %	3.51	−5.68	0.26	16.62	0.39	−3.54	−0.13	5.48	−0.01	−4.06	−0.13	3.57
Sep $	21.77	−25.81	0.06	101.18	−4.49	−31.56	−0.06	46.16	43.74	−65.71	−2.04	83.28
Oct %	3.49	−3.31	−0.11	12.15	0.31	−3.69	−0.20	4.35	1.02	−3.98	−0.28	4.19
Oct $	17.57	−35.83	−0.02	92.80	−7.70	−36.92	−0.23	36.07	28.65	−129.40	−0.65	55.38
Nov %	2.36	−3.78	0.18	12.12	0.17	−3.43	−0.26	3.84	1.63	−1.34	0.29	3.47
Nov $	−1.66	−28.96	0.02	43.75	−9.92	−37.92	−0.48	28.17	43.43	−41.22	2.25	108.29
Dec %	2.96	−3.67	0.20	12.30	0.22	−3.80	−0.35	4.16	−0.57	−6.02	0.20	4.12
Dec $	−2.36	−47.53	0.03	38.04	−6.39	−34.53	−0.58	33.16	85.84	−56.81	1.87	106.81

Table 6, p. 23. Discount and Low Deviation ETFs.

	January–April 2013				May–August 2013			
	Disc.	Low dev.	Diff.	p	Disc.	Low dev.	Diff.	p
Size ($ mln)	904.89	3,795.31	−2,890.42	< .0001	908.31	3,694.40	−2,786.09	< .0001
Volume (shares)	135,797	587,130	−451,332	< .0001	181,785	673,265	−491,480	.002
ETF turnover (%)	85.28	40.17	45.10	.137	58.92	39.27	19.65	.351
ETF bid-ask (%)	0.57	0.16	0.41	.000	0.77	0.21	0.56	< .0001
Average rating	8.00	7.33	0.67	.532	8.20	7.35	0.85	.382
Average effective duration	6.51	5.15	1.36	.219	6.44	5.05	1.39	.164
Average coupon	4.61	4.62	−0.01	.986	4.48	4.55	−0.07	.874

Table 7, p. 26. Treatment and control bonds. Signs are as printed.

	Non-LQD	LQD	Difference	t-value	p-value
ETF pressure	0.277	0.458	0.181	2.50	.013
Amount outstanding ($ mln)	415.832	1,122.307	706.475	15.30	.000
Rating	7.415	7.326	−0.089	−0.42	.674
Time to maturity	15.273	16.488	1.215	1.14	.269
Coupon	5.224	5.441	0.216	1.14	.254
MF pressure	1.082	0.757	−0.326	0.95	.341
Amihud	0.011	0.009	−0.002	1.64	.101
Turnover	0.049	0.041	−0.008	1.07	.285

Table 8, p. 27. LQD threshold. Issuer fixed effects, clustered by issuer.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
LQD bond	0.113**	0.085**	0.112***	0.102**	0.052	0.029	−0.003	0.006	0.004	0.035
	(2.67)	(2.20)	(3.47)	(2.51)	(1.09)	(0.80)	(−0.08)	(0.18)	(0.09)	(0.94)
MF pressure	−0.023	−0.038	−0.016	0.000	−0.006	0.016	−0.004	0.006	0.011	0.001
	(−0.64)	(−1.47)	(−0.67)	(0.01)	(−0.15)	(0.57)	(−0.17)	(0.26)	(0.35)	(0.05)
Time to maturity	0.001	0.000	0.001	−0.002	0.000	0.003	0.004	0.011***	0.015***	0.018***
	(0.23)	(0.07)	(0.46)	(−0.48)	(0.09)	(0.92)	(1.46)	(4.20)	(3.82)	(5.05)
Coupon	0.004	−0.025	−0.004	−0.014	−0.002	−0.001	−0.014	0.007	0.002	−0.016
	(0.23)	(−1.46)	(−0.30)	(−0.91)	(−0.11)	(−0.03)	(−0.93)	(0.53)	(0.10)	(−1.02)
Amihud	3.971	−0.293	−0.595	6.049	11.018***	11.642***	5.639**	9.049***	7.818**	7.772**
	(0.92)	(−0.08)	(−0.18)	(1.58)	(2.94)	(3.58)	(2.13)	(2.85)	(2.24)	(2.60)
May spread	−0.388**	−0.176	−0.280**	−0.229	−0.432***	−0.324**	−0.239**	−0.374***	−0.415***	−0.478***
	(−2.41)	(−1.32)	(−2.42)	(−1.63)	(−2.73)	(−2.57)	(−2.10)	(−4.06)	(−3.07)	(−3.08)
Observations	151	164	165	158	156	160	171	170	167	164
Adj. R^2	.852	.876	.904	.897	.887	.894	.837	.883	.849	.897

Table 9, p. 29. Trade-based pressure. Issuer fixed effects, clustered by issuer.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
ETF pressure	0.011	0.039**	0.041***	0.045**	0.045**	0.038**	0.014	0.020	−0.007	0.003
	(0.77)	(2.52)	(2.67)	(2.30)	(2.25)	(2.05)	(0.92)	(1.18)	(−0.38)	(0.20)
MF pressure	−0.018***	−0.012*	−0.003	0.004	−0.005	−0.003	0.010	0.008	0.008	0.007
	(−2.73)	(−1.76)	(−0.38)	(0.50)	(−0.65)	(−0.32)	(1.60)	(1.15)	(1.01)	(0.96)
Time to maturity	−0.002*	−0.000	−0.001	0.000	0.002	0.006***	0.004***	0.008***	0.008***	0.011***
	(−1.73)	(−0.01)	(−1.01)	(0.05)	(0.80)	(3.51)	(3.08)	(5.47)	(4.89)	(9.26)
Coupon	−0.025***	−0.026***	−0.021***	−0.034***	−0.030***	−0.036***	−0.033***	−0.034***	−0.037***	−0.040***
	(−3.64)	(−3.73)	(−3.12)	(−3.98)	(−3.41)	(−4.59)	(−5.25)	(−4.70)	(−4.54)	(−6.77)
log(size)	0.086***	0.103***	0.069***	0.105***	0.071**	0.086***	0.046**	0.044*	0.007	0.023
	(3.59)	(4.17)	(2.81)	(3.51)	(2.20)	(3.02)	(1.98)	(1.70)	(0.24)	(1.10)
Option dummy	0.057	0.058	0.080	0.054	0.022	−0.054	0.034	0.057	0.049	−0.015
	(0.70)	(0.64)	(0.93)	(0.48)	(0.20)	(−0.51)	(0.39)	(0.63)	(0.49)	(−0.19)
Amihud	1.118	0.352	1.848	2.474	5.798**	5.028**	2.661	5.542**	4.894**	3.491*
	(0.57)	(0.17)	(0.92)	(0.89)	(2.06)	(2.03)	(1.38)	(2.55)	(2.03)	(1.96)
May spread	0.039	0.025	0.024	−0.008	−0.182***	−0.132***	−0.102***	−0.162***	−0.143***	−0.180***
	(1.46)	(0.85)	(0.80)	(−0.21)	(−4.65)	(−3.78)	(−3.44)	(−4.89)	(−3.93)	(−6.65)
Observations	1,138	1,204	1,200	1,133	999	1,159	1,247	1,197	1,155	1,169
Adj. R^2	.759	.646	.737	.655	.720	.754	.813	.796	.798	.888

Table 10, p. 30. Hypothetical arbitrage. Issuer fixed effects, clustered by issuer.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
ETFArb	0.097*	0.148***	0.132**	0.144**	0.043	0.091	0.030	−0.046	−0.112*	−0.052
	(1.72)	(2.70)	(2.43)	(2.15)	(0.57)	(1.33)	(0.52)	(−0.76)	(−1.69)	(−0.87)
Time to maturity	−0.002	0.001	−0.002	−0.002	0.000	0.007***	0.006***	0.011***	0.009***	0.013***
	(−1.49)	(1.09)	(−1.31)	(−1.36)	(0.22)	(4.46)	(5.12)	(7.98)	(6.16)	(9.86)
Coupon	−0.016**	−0.030***	−0.036***	−0.049***	−0.049***	−0.063***	−0.044***	−0.043***	−0.048***	−0.049***
	(−2.24)	(−4.12)	(−5.27)	(−5.62)	(−4.91)	(−7.10)	(−6.26)	(−5.34)	(−5.45)	(−6.25)
log(size)	0.000	0.013	−0.006	−0.031	0.004	0.056*	0.036	0.084***	0.057*	0.054*
	(0.00)	(0.50)	(−0.22)	(−1.00)	(0.11)	(1.73)	(1.37)	(2.89)	(1.81)	(1.91)
Option dummy	−0.061	−0.016	−0.070	−0.103	−0.145	−0.120	−0.085	−0.072	−0.131	−0.086
	(−0.60)	(−0.16)	(−0.70)	(−0.81)	(−1.08)	(−1.03)	(−0.79)	(−0.64)	(−1.09)	(−0.72)
Amihud	2.642	3.584**	3.393**	4.297*	5.876**	7.505***	5.596***	7.061***	6.774***	4.862***
	(1.56)	(2.07)	(1.99)	(1.86)	(2.44)	(3.59)	(3.33)	(3.72)	(3.27)	(2.65)
May spread	−0.013	−0.075**	0.031	−0.008	−0.105***	−0.155***	−0.142***	−0.191***	−0.083**	−0.151***
	(−0.43)	(−2.51)	(1.08)	(−0.23)	(−2.68)	(−4.44)	(−4.77)	(−5.66)	(−2.23)	(−4.70)
Observations	1,258	1,414	1,408	1,328	1,275	1,333	1,491	1,407	1,343	1391
Adj. R^2	.856	.848	.904	.933	.936	.907	.951	.938	.881	.909

Table A1, p. 33. Monthly quintiles on net creation and redemption.

Period	Quintile	Net ($ mln)	Redemptions ($ mln)	Avg daily dev. (%)	Avg month-end dev. (%)	Avg volatility
May	1	−1,870.00	−3,220.00	0.0596	−0.1751	0.0022
	2	9.48	0.00	−0.0994	−0.1093	0.0032
	3	116.00	0.00	0.5340	0.4654	0.0023
	4	324.00	0.00	0.3081	0.0852	0.0013
	5	3,990.00	−596.00	0.1644	0.0875	0.0009
June	1	−7,480.00	−8,780.00	−0.1925	−0.0532	0.0035
	2	−65.20	−257.00	−0.2749	−0.1834	0.0042
	3	0.00	0.00	−0.2172	−0.4053	0.0075
	4	136.00	−6.46	0.2914	0.2084	0.0028
	5	3,090.00	−323.00	0.2155	0.2208	0.0019
July	1	−1,820.00	−2,200.00	−0.0737	−0.0172	0.0022
	2	−27.70	−27.70	−0.2878	−0.4935	0.0075
	3	7.99	−22.60	−0.0756	−0.1721	0.0041
	4	258.00	−396.00	0.2279	0.2606	0.0032
	5	5,010.00	−814.00	0.3027	0.3498	0.0016
August	1	−4,350.00	−4,990.00	−0.1353	−0.1078	0.0017
	2	−18.60	−35.00	−0.6410	−0.6610	0.0039

Table A1, p. 33. Monthly quintiles on net creation and redemption.

Period	Quintile	Net ($ mln)	Redemptions ($ mln)	Avg daily dev. (%)	Avg month-end dev. (%)	Avg volatility
	3	1.15	0.00	−0.2357	−0.4126	0.0058
	4	136.00	−96.30	0.3592	0.3057	0.0078
	5	2,630.00	−300.00	0.2616	0.2590	0.0009

Table B1, p. 34. Key rows only.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
A. Match on Roll										
LQD bond	0.138***	0.081**	0.104***	0.107***	0.071	0.078**	0.010	0.052	0.038	0.065
	(3.54)	(2.35)	(3.58)	(3.01)	(1.31)	(2.10)	(0.30)	(1.54)	(0.75)	(1.68)
MF pressure	0.025	−0.019	0.009	0.039*	0.022	0.051*	0.006	0.017	0.013	0.015
	(0.79)	(−0.81)	(0.41)	(1.85)	(0.55)	(1.84)	(0.24)	(0.72)	(0.38)	(0.62)
Roll	12.158	4.941	−6.103	10.960	21.051	28.277**	5.815	16.781	9.815	8.274
	(0.90)	(0.45)	(−0.67)	(1.05)	(1.66)	(2.67)	(0.68)	(1.58)	(0.78)	(0.86)
Observations	151	161	160	152	148	158	168	167	161	161
Adj. R^2	.901	.915	.946	.936	.899	.915	.836	.874	.831	.896
B. Match on IRC										
LQD bond	0.094	0.104***	0.083**	0.099***	0.061	0.096***	0.010	0.029	−0.036	0.003
	(1.62)	(3.88)	(2.58)	(2.94)	(1.29)	(2.83)	(0.30)	(0.77)	(−0.85)	(0.07)
MF pressure	0.049	0.062**	0.021	0.056**	0.015	0.052*	−0.004	−0.022	−0.024	0.031
	(1.32)	(2.76)	(0.76)	(2.09)	(0.36)	(1.94)	(−0.15)	(−0.80)	(−0.78)	(0.93)
IRC	20.545	36.919**	22.429	40.956	41.939	42.573*	5.987	31.410	16.383	24.938
	(0.48)	(2.11)	(0.97)	(1.70)	(1.47)	(1.93)	(0.32)	(1.16)	(0.67)	(1.13)
Observations	129	141	140	135	133	137	148	145	145	139
Adj. R^2	.946	.975	.960	.969	.960	.967	.923	.916	.916	.944

Table B2, p. 35. Match on time to maturity. Key rows only.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
A. Amihud										
LQD bond	0.168***	0.095**	0.090**	0.109**	0.062	0.056	−0.018	0.038	0.072	0.065*
	(4.07)	(2.31)	(2.33)	(2.69)	(1.04)	(1.66)	(−0.41)	(1.15)	(1.65)	(1.75)
MF pressure	−0.046	−0.039	−0.054*	0.011	−0.084*	0.010	−0.040	−0.012	0.022	0.008
	(−1.08)	(−1.31)	(−1.77)	(0.38)	(−1.98)	(0.38)	(−1.30)	(−0.49)	(0.64)	(0.27)
Amihud	7.883*	2.382	1.429	9.075**	8.950*	11.378***	5.319	11.565***	9.954***	8.994***
	(2.03)	(0.60)	(0.37)	(2.14)	(1.93)	(3.71)	(1.50)	(3.27)	(2.80)	(2.97)
Observations	130	153	152	147	145	149	165	164	157	152
Adj. R^2	.895	.846	.852	.875	.851	.901	.800	.898	.877	.909
B. Roll (October = November, March = April, as printed)										
LQD bond	0.183***	0.111**	0.085*	0.085*	0.087	0.070	0.002	0.044	0.044	0.096*
	(4.03)	(2.33)	(1.85)	(1.85)	(1.13)	(1.50)	(0.03)	(0.99)	(0.99)	(1.98)
MF pressure	−0.055	−0.073*	−0.050	−0.050	−0.145**	0.024	−0.078*	−0.013	−0.013	−0.021
	(−1.25)	(−1.79)	(−1.31)	(−1.31)	(−2.15)	(0.60)	(−1.82)	(−0.36)	(−0.36)	(−0.46)
Roll	22.497	2.284	−18.945	−18.945	3.856	28.158*	11.511	19.915	19.915	18.056
	(1.30)	(0.12)	(−1.05)	(−1.05)	(0.20)	(1.88)	(0.86)	(1.23)	(1.23)	(1.36)
Observations	130	153	152	147	145	149	165	164	157	152
Adj. R^2	.886	.850	.856	.856	.839	.877	.782	.876	.876	.882
C. IRC										
LQD bond	0.202***	0.126**	0.103**	0.111***	0.105	0.062	0.018	0.061	0.080	0.072
	(4.96)	(2.67)	(2.36)	(2.85)	(1.56)	(1.45)	(0.37)	(1.45)	(1.53)	(1.64)
MF pressure	−0.048	−0.071*	−0.055	0.003	−0.122*	0.030	−0.075*	−0.005	0.038	−0.010
	(−1.22)	(−1.77)	(−1.47)	(0.09)	(−1.99)	(0.81)	(−1.83)	(−0.15)	(0.91)	(−0.25)
IRC	73.320***	37.958	40.289	90.417***	127.291**	78.948**	62.310*	69.453**	162.955***	83.189***
	(2.93)	(1.27)	(1.13)	(3.33)	(2.43)	(2.65)	(1.85)	(2.64)	(4.53)	(3.01)
Observations	130	153	152	147	145	149	165	164	157	152
Adj. R^2	.908	.857	.857	.917	.870	.888	.796	.891	.904	.903

Table B3, p. 36. Key rows only.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
A. Roll										
ETF pressure	0.012	0.040**	0.041***	0.044**	0.046**	0.038**	0.014	0.020	−0.007	0.002
	(0.78)	(2.54)	(2.69)	(2.29)	(2.29)	(2.04)	(0.91)	(1.20)	(−0.41)	(0.17)
MF pressure	−0.018***	−0.011*	−0.003	0.005	−0.006	−0.003	0.010	0.008	0.007	0.006
	(−2.71)	(−1.69)	(−0.39)	(0.56)	(−0.66)	(−0.32)	(1.57)	(1.16)	(0.94)	(0.87)
Roll	4.436	16.253**	9.149	16.350*	21.685**	11.446	2.594	10.474	−1.503	−4.159
	(0.69)	(2.28)	(1.25)	(1.87)	(2.11)	(1.37)	(0.43)	(1.40)	(−0.20)	(−0.64)
Observations	1,138	1,204	1,200	1,133	999	1,159	1,247	1,197	1,155	1,169
Adj. R^2	.759	.648	.737	.656	.720	.753	.813	.795	.797	.887
B. IRC										
ETF pressure	0.012	0.034**	0.036**	0.045**	0.043**	0.034*	0.038**	0.023	−0.006	0.000
	(0.83)	(2.29)	(2.55)	(2.49)	(2.44)	(1.94)	(2.39)	(1.32)	(−0.34)	(0.01)
MF pressure	−0.019***	−0.011	−0.002	0.004	−0.002	−0.001	−0.006	−0.011	0.009	0.006
	(−2.85)	(−1.63)	(−0.27)	(0.55)	(−0.19)	(−0.08)	(−0.79)	(−1.34)	(1.12)	(0.92)

Table B3, p. 36. Key rows only.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
IRC	21.684*	4.071	11.947	31.015**	23.908	28.474**	26.148**	29.767**	7.356	0.833
	(1.86)	(0.35)	(1.01)	(1.96)	(1.53)	(2.03)	(2.14)	(2.17)	(0.51)	(0.08)
Observations	1,137	1,261	1,254	1,190	1,157	1,205	1,299	1,258	1,199	1,226
Adj. R^2	.760	.761	.825	.778	.839	.838	.850	.853	.865	.925

Table B4, p. 37. Key rows only.

	Aug 13	Sep 13	Oct 13	Nov 13	Dec 13	Jan 14	Feb 14	Mar 14	Apr 14	May 14
A. Roll										
ETFArb	0.096*	0.138**	0.122**	0.126*	0.031	0.076	0.011	−0.020	−0.085	−0.058
	(1.70)	(2.56)	(2.24)	(1.86)	(0.39)	(1.09)	(0.19)	(−0.34)	(−1.32)	(−0.94)
Roll	3.649	12.104**	4.566	8.574	15.247*	17.271**	7.210	10.016*	7.128	5.320
	(0.67)	(2.50)	(0.78)	(1.33)	(1.88)	(2.41)	(1.48)	(1.76)	(1.23)	(0.87)
Observations	1,258	1,414	1,408	1,328	1,275	1,333	1,491	1,407	1,343	1,391
Adj. R^2	.856	.860	.906	.933	.936	.906	.951	.941	.891	.908
B. IRC										
ETFArb	0.100*	0.133**	0.123**	0.123*	0.024	0.078	0.020	−0.019	−0.084	−0.056
	(1.77)	(2.48)	(2.26)	(1.80)	(0.31)	(1.12)	(0.35)	(−0.32)	(−1.30)	(−0.91)
IRC	20.791**	13.899	11.651	3.490	23.011	34.178***	26.012***	27.630***	13.675	12.185
	(1.98)	(1.48)	(1.21)	(0.29)	(1.61)	(2.79)	(2.72)	(2.70)	(1.19)	(1.10)
Observations	1,258	1,414	1,408	1,328	1,275	1,333	1,491	1,407	1,343	1,391
Adj. R^2	.856	.859	.906	.933	.936	.906	.951	.942	.891	.908

6.1 Figure values, read by eye

[INFERRED: the paper prints no figure values, so these are my approximate readings. Use them only as rough checks**]**

- Figure 2 panel C, Low Deviation redemptions, $ billions: May 2.6, June 6.2, July 2.2, August 3.4.
- Figure 2 panel C, Discount redemptions, $ billions: May 0.35, June 1.7, July 1.1, August 1.0.
- Figure 2 panel C, Premium redemptions, $ billions: May 0.9, June 1.5, July 0.2, August 1.0.
- Figure 2 panel B, total assets: Low Deviation near $110 billion, Discount near $20 to $25 billion, Premium near $30 to $34 billion.
- Figure 3 panel A, LQD minus non-LQD spread: about −0.025 in January 2013, a peak near 0.103 in August 2013, and near zero by July 2014.

6.2 Quoted figures in the text, checked against the tables

Text claim	Page	Arithmetic	Verdict
ETF flow volatility 1.9 times mutual funds	12	7.13/3.75 = 1.90. 7.13/3.69 = 1.93.	Matches
1.4 points above, a 37% increase	2	1.4/3.75 = 37.3%. Table 2 column 2 is 1.454.	Matches when rounded
Minimum 40% (18.4%) increase	13	1.454/3.75 = 38.8%. 0.9/4.88 = 18.4%.	40% does not match 37% on p. 2
Interaction gives 0.725% and 0.790%	14	0.484 + 0.241 = 0.725. 0.524 + 0.266 = 0.790.	Matches
A 37.5% and 41.1% increase over mean flow	14	0.725/1.92 = 37.8%. 0.790/1.92 = 41.1%.	First is off by 0.3
ETF outflow multiples of 9.3, 9.1, 7.6	19	7.82/0.84, 7.66/0.84, 6.37/0.84.	Matches
Volume up 34% and 15%	21	181,785/135,797, 673,265/587,130.	Matches
11.3 and 8.5 bps, as 10.1% and 7.6% of the spread	26	Both imply a pre-episode mean LQD spread near 1.12%.	Consistent
4.0 bps in the month after	5	Table 9 September is 0.039, so 3.9 bps.	Text rounds up
14.8 bps	5, 28	Table 10 September is 0.148.	Matches
20.35 bps at peak	5, 28	4.5 × 4.52 = 20.34.	Matches within rounding
Average monthly ETF return is 22 bps	28	Table 1 mean is 0.20%.	Does not match

Text claim	Page	Arithmetic	Verdict
Median ETF 3%, active 15%, index 43% liquid	3	$0.91 + 2.13$, $3.77 + 11.24$, $1.33 + 42.06$.	Matches
Index median 42.6% Treasuries, 26.23% corporate	16	Table 1 has 42.06 and 26.38.	Does not match
Table 4 p-values	18	0.946, $t = 15.13$ gives $\lvert 1 - \beta \rvert / \text{se} = 0.86$, $p \approx .39$. 0.872, $t = 15.75$ gives 2.31, $p \approx .02$.	Matches

7 Label register

Labels sit next to each item throughout this document. This section collects the ones that matter. The label counts in this file are: 180 STATED, 43 INFERRED, 67 MISSING and 9 MINE. These counts include the four examples in the reading guide.

7.1 Gaps ranked by how much they could move Table 8, 9 or 10

Ranked gaps and conflicts

#	Gap	What the paper says	Proposed default
1	Control bond rule	Closest in Amihud (p. 25, Table 7 note). Largest below the threshold (Table 8 note). These conflict.	**[MISSING**. Proposal: closest May Amihud, because the body text and Table B1 match on liquidity. Also run the largest-below rule, and report which is closer to the targets**]**
2	Which bonds count in Table 9	"In baskets exchanged with APs" (p. 26). Table 9 has fewer bonds than Table 10.	**[MISSING**. Proposal: require at least one ETF holding change**]**
3	Spread construction	Volume-weighted yield over the matched Treasury rate. No source for yields or Treasury rates.	**[MISSING**. Proposal: trade-level yield minus H.15 interpolated, then par-weighted mean**]**
4	Winsorising scope	1% and 99%, no scope given.	**[MISSING**. Proposal: by month over all filtered bonds, before differencing**]**
5	May-level Amihud window	"May levels". Onset falls inside May.	**[MISSING**. Proposal: 1 to 21 May**]**
6	Whether treatment bonds must be held by LQD	Eligibility language, with no holding check.	**[MISSING**. Proposal: eligibility first, holding as a variant**]**
7	Index issuer rules	Cited by a link only.	**[MISSING**. Proposal: size and maturity rules, then add the 2013 issuer rules if found**]**
8	Amount-change window	No dates given.	**[MISSING**. Proposal: 22 May 2013 to 31 May 2014**]**
9	Rating aggregation	"Most recent" from three agencies.	**[MISSING**. Proposal: mean**]**
10	Quarterly flow, new positions (Table 4)	Not defined.	**[MISSING**. Proposal: Eq. (1) between reports, and drop new positions**]**
11	Table 7 turnover	Not defined.	**[MISSING**. Proposal: monthly par turnover**]**
12	Summary statistics of ETFArb	None reported.	**[MISSING**. Proposal: compute and publish them**]**

7.2 Internal conflicts in the paper

These are errors or tensions inside the paper. I checked each against the page image. None changes the headline result by itself. Items 1 and 2 change what you build. Item 3 means you cannot use those columns as targets.

1. **Control rule.** See the first row of the ranked table in Section 7.1. **[STATED p. 25, p. 26, p. 27]**
2. **Size control in Table 8.** The p. 26 text excludes it and the table shows no size row. The Table 8 note still lists the log of amount outstanding. I follow the text.
3. **Table B2 panel B has duplicated columns.** October equals November, and March equals April, in every coefficient and t-statistic. Bond counts differ (152 and 147, 164 and 157). The authors' supplementary file has the same numbers, so this is not a typesetting error. Treat those four columns as unreliable.
4. **Trading days.** "Last three full trading days" (p. 8, p. 25) against "last 3 business days" (Tables 8–10 notes).
5. **No negative reversal claim.** P. 31 says the pattern of negative significant coefficients after September appears in no regression. Table 10 April 2014 is -0.112, $t = -1.69$, significant at 10%. It is a single month at the weakest star level.
6. **Table 7 signs.** MF pressure, Amihud and turnover have negative differences with positive t-values. Time to maturity and coupon both show $t = 1.14$, with p-values .269 and .254.
7. **Amihud scale.** Table 1 mean is 0.73, and Table 7 means are 0.011 and 0.009.
8. **Figure 2 dates.** P. 21 says January 2013 to June 2014. The caption and the axis say January 2012 to December 2014.
9. **Figure 1 axes.** The caption swaps the axes of panel B. The figure's axis titles match the p. 6 text.
10. **Redemptions share.** P. 21 says Low Deviation ETFs account for "nearly all" redemptions. Reading the bars of panel C by eye gives about two-thirds. P. 4 says "most". **[INFERRED**: from my reading of the bars, so this is approximate**]**
11. **Table 6 volume.** The row says shares and the note says dollars.
12. **Notes.** The Table 9 note defines the option dummy as "equal to zero if a bond has option features and zero otherwise". The Table 10 note says it repeats Table 7, and it should say Table 8 or 9.
13. **LQD history.** Fn. 11 says 18-year history. Fn. 26 says 126-month history through June 2020.
14. **Bond counts across proxies.** Table B3 panel A has the same counts as Table 9. Table B4 panels A and B have the same counts as Table 10. The liquidity proxies should have different missing patterns. Table B3 panel B counts do differ. **[INFERRED**: the identical counts suggest a shared sample or missing values filled in, which the paper does not explain**]**
15. **December 2013 count.** Table 9 has 999 bonds in December, against 1,133 to 1,247 in other months. Table B3 panel B has 1,157 in that month. The paper gives no reason.

7.3 Errors in the earlier notes

I checked `E_local_documents.md`, Section 4.4 (lines 117–136). **I found no wrong number. The notes are incomplete in five places.**

1. **"8.5 to 11.3 bps".** The note gives 11.3 in August and 8.5 in September, which is what the paper says on p. 26. But October (11.2) and November (10.2) are also significant. So 8.5 is the low of a four-month range, not the end of a decay. The path does not fall steadily. It is also not a confidence interval.
2. **"Episode runs 21 May to 18 September".** Only the LQD share statistic uses those dates. The paper's onset is 22 May, and the pressure measures stop at the end of August.
3. **"4.0 bps a month after the episode".** The text on p. 5 says this. Table 9 gives 0.039, so 3.9 bps. Use 3.9.
4. **"1.4 percentage points, 37%".** This follows p. 2. Table 2 column 2 is 1.454, and p. 13 says a minimum of 40%. The p. 2 phrase "lagged fund characteristics" also does not fit column 2, whose controls are sample averages. The lagged panel, column 4, gives 2.155.
5. **"One for one".** This holds only for outflows and only with controls: 0.946, $p = .39$. Without controls the slope is 0.872, and the paper's table rejects one at $p = .02$. Inflows are 0.411. The note's data list also omits Morningstar, Bloomberg and Mergent FISD.

8 Threats to validity

8.1 As the paper handles them

Threat	Paper's response	Label
Investors sell the funds most exposed to rates. Managers sell the weakest bonds.	Issuer fixed effects compare bonds with one credit risk.	**[STATED p. 4, p. 23]**
Selection into ETFs.	Only bonds in ETF portfolios enter.	**[STATED p. 4, p. 24]**

Threat	Paper's response	Label
Term structure differences.	Spreads against maturity-matched Treasury rates.	**[STATED p. 4, p. 25]**
Selling due to the 3-year index rule.	At least 4 years to maturity in setting 1.	**[STATED p. 4, p. 25]**
Selling due to the 1-year index rule.	Drop bonds under 1 year and 4 months.	**[STATED p. 11]**
Forced selling on a downgrade.	Drop bonds crossing investment grade, by any one agency.	**[STATED p. 11, fn. 15]**
Treatment and control differ.	Table 7 shows no significant gap in rating, maturity, coupon, MF pressure, Amihud or turnover.	**[STATED p. 26]**
Mutual fund selling drives the result.	MF pressure control. It is insignificant in Table 8 and negative in early Table 9 months.	**[STATED p. 26, p. 28]**
Manager discretion in baskets.	Hypothetical measure with pre-episode weights.	**[STATED p. 28]**
Mechanical link in scaling.	Scale by amount outstanding, not dollar volume, citing Wardlaw (2020).	**[STATED p. 12]**
Choice of liquidity proxy or matching variable.	Tables B1–B4.	**[STATED p. 26, p. 28]**
Price discovery.	Coefficient path and Figure 3 panel B.	**[STATED p. 5, p. 31]**
Timing of AP sales unobserved.	Acknowledged. Dealer inventory half-life of 4 to 6 weeks or over a month supports a slow effect.	**[STATED fn. 7, p. 28]**
One episode.	Discussed through a contrast with COVID-19, where APs let discounts persist and the Fed intervened.	**[STATED p. 6, p. 32]**

8.2 My own concerns

All items below are **[MINE**: my analysis, using the paper's own numbers. They are not claims by the authors**]**

1. **Table 8 may partly measure bond size.** This is the most important concern. Treatment bonds average $1,122 million and controls $416 million. So this is not a narrow comparison at the threshold. The log gap is 0.99. The Table 9 slopes on log size in August to November are 0.086, 0.103, 0.069 and 0.105. Those slopes alone imply 7 to 10 bps across that gap, close to the LQD coefficients of 8.5 to 11.3 bps. Table 8 omits size. But the evidence is mixed: the Table 10 size slopes are near zero in those months. And in Table 9 the size slope may itself capture LQD eligibility.
2. **The measured pressure gap does not explain the LQD effect.** Table 7 shows a pressure gap of 0.181. At the Table 9 peak slope of 0.045, that predicts 0.8 bps. Table 8 shows 8.5 to 11.3 bps, more than ten times larger. Either the pressure measure has large errors, or the LQD dummy captures something else, such as size or LQD-specific trading.
3. **No test of pre-trends.** Panel A of Figure 3 shows the LQD gap at about −2.5 bps in January and February 2013, rising to about +2 to +3 bps in March and April, before the onset. These are my readings of the chart. The authors attribute early moves to market jitters (p. 23). A placebo regression with a pre-May baseline would test this. The May baseline may also already contain some of the effect, which would understate it.
4. **Repeated cross-sections.** A bond enters a month only if it trades in the last three full days. So the sample changes month to month, and trading itself may respond to pressure. A fading coefficient could partly reflect a change in sample. A balanced panel would test this.
5. **Many tests.** Tables 8–10 hold 30 monthly tests. No correction for multiple testing appears. Most significant t-statistics lie between 2.05 and 2.70. The months overlap in sample and baseline, so the tests are not independent.
6. **Few clusters in Table 8.** I infer about 75 to 85 issuers. With issuer effects absorbing about half of the 151 to 171 observations, little variation remains. Adjusted R-squared of .837 to .904 may mostly reflect the fixed effects. Wild cluster bootstrap inference would be a useful check.
7. **Regression to the mean.** The May spread sits on both sides of the equation. Its coefficient is often large and negative. If May spreads are noisy, the control absorbs noise and may bias other slopes.
8. **Yield spread is not bond-specific in rate risk.** Treasury yields rose sharply. The maturity match leaves convexity, coupon and call effects in the spread. Table 8 treated and control bonds average 15 to 16 years, so these effects can be large. An option-adjusted measure would be cleaner. This also weakens the claim that the difference is "in excess of" term structure changes.
9. **The channel is inferred, not observed.** The paper does not show that APs sold these bonds. Bond-level customer selling in TRACE, by month and dealer direction, could test it directly.

10. **Holdings timing.** Pressure spans report dates that can sit well outside May to August. Index rebalancing and ordinary trading enter ΔH.
11. **The August regression is not out of sample.** The ETF measures run to 31 August. The August spread is measured in the same days.
12. **Reversal is confounded with the market.** Spreads fall for all bonds after September (Figure 3 panel B). Reversal coincides with a market-wide rally and the 18 September Fed decision. This is the authors' setting, so it cannot be separated within one episode.
13. **External validity.** This is one rate-driven episode in a young market. ETF ownership averaged about 2% per bond (Table 1). ETF assets have since more than doubled (p. 6, p. 32). AP behaviour differed in March 2020 (p. 6, p. 32). Rules on ETF baskets may also have changed since 2013, which I did not check. A credit-driven episode, or a calm period, may behave differently.

9 Adaptation to TRACE positioning, ETF holdings or RFQ flow

This whole section is **[MINE**: my view, not the paper's, and it is not a study design**]**

What would carry over.

- **Same-issuer comparison.** Issuer fixed effects remove issuer credit news from any bond-level flow measure. Any TRACE or RFQ pressure signal needs this control.
- **Actual against hypothetical exposure.** The trade-based and weight-based pair is a useful template. Weights times flows avoid manager choice. They work wherever holdings and shares outstanding exist.
- **Creation and redemption dollars as flow.** Eq. (4) needs only NAV and shares. It extends to any ETF with daily data.
- **The reversal test.** A pressure signal should predict a move and then its reversal over months. So a 10-day label sees only the first part.
- **Scaling by amount outstanding.** This avoids the mechanical link with volume.
- **Last-three-days spread.** It is a simple way to get month-end spreads for illiquid bonds.
- **The deviation sort.** The result that the largest ETFs carry the redemptions suggests weighting flow by the ETF's arbitrage activity.

What would not carry over.

- **The event design.** The design needs one large shock. A continuous signal needs a panel with time effects and a different inference approach.
- **The sample size with 15 issuers.** Issuer effects with 15 issuers leave too few clusters for the paper's inference, and no issuer universe to generalise from.
- **The LQD threshold.** The $750 million rule belongs to the 2013 index. Any current use needs today's rule book. Most of the 15 issuers may have few bonds below it.
- **Quarterly holdings.** These are too slow for RFQ-scale signals. N-PORT and daily sponsor files change what is possible, but not for 2013.
- **Causal reading of RFQ flow.** The paper's flows come from a fund's redemptions, which are plausibly outside the bond's own news. RFQ flow is chosen bond by bond, so it may carry bond-specific information. The hypothetical measure has no direct RFQ analogue.
- **Dealer direction.** The paper uses no buy or sell side from TRACE and no dealer data. So it gives no guidance on signing flow.
- **Magnitudes.** The basis point effects belong to 2013 ownership levels and that one episode.

10 Acceptance checks, cheapest first

The tolerances are **[MISSING**. Proposal: within 1 standard error for key coefficients, within 5% for bond counts and within 10% for fund counts. Same star level or one level away. The paper sets no tolerance**]**

1. **Code test with synthetic data.** Plant a known β_1 in a fake panel with issuer effects. Recover it. Check that flipping the sign flips β_1. Cost: no data needed.
2. **Arithmetic of the targets.** Reproduce the table of quoted figures in Section 6. Cost: no data needed.
3. **LQD share count.** Shares outstanding fall 22.9% from 21 May to 18 September 2013. Average daily deviation is 0.1 bps, and the worst end-of-day discount is 87 bps (p. 23). Cost: public sponsor data.
4. **Table 5.** Reproduce the 10th percentile flows and the multiples 9.3, 9.1 and 7.6 (p. 19). Cost: CRSP monthly only.

5. **Table 1 panel A.** Match means and medians. Check 7.13 against 3.75 for flow volatility. Cost: CRSP and Morningstar.
6. **Tables 2 and 3.** Match the ETF coefficient in columns 1 and 2 of Table 2, then the interaction in Table 3 columns 1 and 4. Cost: CRSP and Morningstar.
7. **Redemptions tie out.** Table A1 redemptions by month sum to about $3.8, $9.4, $3.5 and $5.4 billion. Figure 2 panel C by eye gives the same totals. **[INFERRED**: I summed the Table A1 rows and read the bars**]** Cost: daily ETF data.
8. **Table 6 and Figure 2.** Low Deviation means near 3,795 and Discount near 905 in pre-Tantrum size. Cost: daily ETF data.
9. **Table 4, columns without controls.** ETF outflow 0.872 and $p = .02$. Inflow 0.362. Cost: holdings.
10. **Bond universe counts.** 10,131 bonds, 67%, 96% and 48%, 1,491 issuers and 68. Cost: holdings and FISD.
11. **Table 4 with controls.** ETF outflow 0.946 and $p = .39$. Cost: adds full TRACE.
12. **Scale check.** May Amihud means near 0.011 and 0.009 in the Table 7 sample, and Table 1B Amihud near 0.73. If you get these, your scale matches the paper's two scales. Cost: TRACE 2013.
13. **Table 10, then Table 9.** Run these before Table 8 because they have no matching step. Targets: N, then the sign pattern. Table 10 is positive and significant August to November. Table 9 is significant September to January. Cost: all of the event data.
14. **Table 7, then Table 8.** Build both control rules. The right rule should give 151 to 171 bonds, the Table 7 balance and 0.113, 0.085, 0.112, 0.102 for August to November. Cost: event data plus the index rules.
15. **Appendix B.** Compare with Tables B1, B3 and B4. Ignore the four duplicated columns in B2 panel B.
16. **Falsification beyond the paper.** **[MINE**: run the same regressions with a February 2013 baseline and April 2013 as the outcome. The ETF proxies should have zero slope. A non-zero slope points to the pre-trend concern in Section 8**]**

What failure means. A gap in checks 1 to 12 is a build error. Fix it before going further. A gap in 13 or 14 may be a build error, or the result of a Section 7 choice. Vary those choices one at a time, in ranked order, and log which ones close the gap.